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Brain-inspired memory architecture for condition monitoring based on hippocampal-neocortical complementary learning

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ABSTRACT

This paper tackles the critical challenge of catastrophic forgetting and inefficient learning in artificial intelligence models processing continuous, non-stationary data streams. Inspired by neurobiological mechanisms, specifically the Complementary Learning Systems (CLSs) theory involving the hippocampus and neocortex, we propose a novel brain-inspired biomimetic memory system. The core innovation integrates dimensionality reduction techniques—covariance decomposition and low-dimensional mapping—for efficient feature extraction from long spatiotemporal-scale information flows, with a biomimetic learning/forgetting mechanism grounded in CLS principles. Evaluated on real-world power plant operational data, the proposed system demonstrates robust performance against noise and fluctuations, validating the effectiveness of the bionic learning/forgetting mechanism.

1. Introduction

Modern industrial systems, particularly safety-critical complex facilities like rail transit, rely heavily on condition monitoring and alarm analysis to ensure safe and stable operation. Rail transit systems feature intricate structures and prolonged operational cycles, generating continuous and non-stationary data streams during operation. However, a key challenge arises: actual operational data undergoes slow, continuous drift over time due to system aging, equipment wear, maintenance adjustments, or environmental changes (i.e., slow time-varying effects) [1]. This dynamic nature poses severe problems for static monitoring models trained on historical data: post-deployment models fail to adapt effectively to new data distributions (unrelated to model generalization ability [2]), leading to high false alarm rates and missed detection risks [3]. Specifically, when the statistical properties of monitoring data (e.g., mean, variance, correlation) gradually deviate from the training set distribution over time, models based on fixed thresholds or fixed training sets misidentify this “drift” as an anomaly, generating numerous false alarms. Concurrently, the model’s sensitivity to genuine early-stage, minor faults may also diminish.

Addressing these challenges necessitates models with Continual Learning (CL) or Lifelong Learning (LL) capabilities [4]. This means models must continuously acquire new knowledge and adapt to new environments throughout their lifecycle while retaining competence on previously learned tasks (e.g., known fault pattern recognition). The goal of continual learning is to endow models with adaptive intelligence akin to biological systems, enabling them to autonomously process new information within non-stationary data streams [4]. Oussadek introduced DOLFIN, a methodology for Federated Continual Learning (FCL) that achieves an optimal balance between stability and plasticity by integrating orthogonal low-rank adapters with a Dual Gradient Projection Memory (Dual GPM). This approach was found to exceed multiple strong baselines in terms of final average accuracy [5]. For Large Language Models (LLMs), Lin et al. [6] proposed sparse memory finetuning, which has been shown to drastically reduce catastrophic forgetting (experiencing only an 11 % drop in performance on prior tasks) by updating only the memory slots most relevant to new knowledge. Addressing Domain Incremental Learning (DIL), Paeedeh et al. [7] developed CONEC-LORA, a framework that explicitly consolidates task-shared and domain-specific knowledge via LoRA modules, further utilizing a stochastic classifier and an auxiliary network for enhanced generalization. Furthermore, the EdgeSync framework by Dong et al. [8] tackles data drift in real-time video analytics by implementing an efficient adaptive continuous learning process on edge devices, utilising a novel sample filtering method and a dynamic training manager to accelerate model updates and improve accuracy.

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Mandivarapu presented COLA, a novel CL approach for LLMs that eliminates the need for data replay by employing a lifelong autoencoder to efficiently compress and retrieve task-specific adapter (LoRA) weights, thereby mitigating forgetting while significantly reducing computational overhead [9]. Current continual learning research primarily revolves around three paradigms:

1) Dynamic model based CL. Aims to transform static models into dynamically updatable structures, enhancing adaptability for large, complex systems (e.g., rail transit equipment) [10]. The performance of traditional models (statistical or machine learning) is strictly limited by their training set, essentially representing feature extraction from specific historical data. Dynamic models, however, can adaptively update based on real-time data or specific problems, maintaining alignment with the current system state, and potentially improving performance through feature selection [11].

2) Data flow based CL. Focuses on the efficient utilization of existing data, sample features, or labels. Transfer learning is a representative method, leveraging knowledge from a source domain to assist learning in a target domain, mitigating data scarcity issues (especially for fault data) in the target domain [12,13]. However, validating the effectiveness of transferred models in the target domain often relies on sufficient target domain data.

3) Holistic framework based continual learning. Focuses on constructing more general, autonomous learning frameworks rather than targeting specific processes or single models. These methods integrate multi-task learning, meta-learning, and transfer learning to design learning logic and knowledge management strategies at the framework level [14,15].

Despite its significant potential, the practical application of continual learning in condition monitoring faces notable challenges.

1) Data bottleneck. Operational data, particularly fault samples reflecting diverse fault modes, is insufficiently accumulated and difficult to obtain. This data scarcity severely constrains the effectiveness of methods like transfer learning that rely on extensive data for validation.

2) Learning logic and interpretability. How to design a learning/forgetting mechanism that aligns with engineering practice and is interpretable? How to define and quantify new knowledge, obsolete knowledge, or knowledge to be forgotten? These core questions remain inadequately addressed.

3) Validation difficulty. Assessing the performance evolution and stability of models during continual learning lacks standardized validation methods.

Addressing the specific needs of long-term monitoring and the shortcomings of existing continual learning methods, this paper proposes an innovative solution inspired by neurobiology. The inspiration stems from the Complementary Learning Systems (CLSs) theory of the human brain [16]. CLS theory elucidates the synergistic role of the Hippocampus and Neocortex in memory formation: the Hippocampus rapidly learns new information (short-term memory), while the Neocortex transforms crucial information into long-term stable storage (long-term memory) through a slow consolidation process, effectively preventing new knowledge from overwriting old knowledge (catastrophic forgetting). This biological mechanism provides a highly instructive blueprint for solving the adaptability problem of machine learning models in non-stationary data streams.

The primary objective of this study is to map and apply the core principles of CLS theory to the field of industrial process monitoring, specifically targeting the challenges of long-term condition monitoring. We propose a brain-inspired memory architecture and construct a continual learning model named CLS-PCA based on Principal Component Analysis (PCA). The core innovations of this model lie in:

1) Bio-inspired learning/forgetting mechanism. Explicitly defining the model's learning (simulating hippocampal rapid acquisition) and forgetting (simulating neocortical selective consolidation) processes, and concretizing their operational targets as the dynamic update (addition/deletion) of the model's training dataset (long-term memory).

2) Gating unit design. Drawing on the gating concept of Long Short-Term Memory (LSTM) networks, we design a learning gate (L_t) and a forgetting gate (F_t), coupled with a learning factor ($S_l(t)$) and a forgetting factor ($S_f(t)$). These components precisely control the timing and conditions for integrating new data into the training set (knowledge consolidation) and removing old data from the training set (knowledge obsolescence), ensuring the stability and efficiency of the update process.

3) Adaptive threshold calculation. Based on the dynamically updated training set, the monitoring thresholds (T^2 and Q statistic thresholds) for the PCA model are recalculated in real-time. This enables the model to automatically adapt to the slow time-varying nature of the data distribution, significantly reducing false alarms caused by distribution shifts while maintaining monitoring capability for genuine faults.

The subsequent sections of this paper are organized as follows: Section 2 details the CLS theory and its mapping definitions to machine learning models. Section 3 provides an in-depth analysis of the limitations of traditional PCA and Dynamic PCA (DPCA) in long-term monitoring, presents the CLS-PCA model structure, gating mechanism design, and adaptive threshold calculation method. Section 4 validates the performance of CLS-PCA using actual long-term operational data from a power plant and simulated fault data. Section 5 summarizes the work and outlines future research directions.

2. Complementary learning systems: bridging neurobiology and machine continual learning

2.1. Theoretical foundation of CLS

The core challenge of continual learning capability is how to enable models to absorb new knowledge while avoiding catastrophic forgetting of previously learned knowledge. This challenge has been profoundly explained in neuroscience through the CLS theory [16]. CLS theory reveals the synergistic mechanism of two key subsystems in the human brain—the Hippocampus and the Neocortex—in memory formation and consolidation.

Hippocampus: Functions as a rapid learning engine, responsible for the instantaneous encoding (episodic encoding) and short-term storage (short-term retention) of new information. Its high plasticity allows rapid adaptation to new experiences, but its storage capacity is limited and susceptible to interference.

Neocortex: Functions as a distributed knowledge repository, responsible for transforming key information from the Hippocampus into long-term stable representations (long-term stable representations) through a slow consolidation process. Its low plasticity ensures knowledge persistence but results in slower learning speeds.

Core mechanism: When new information is input (denoted as $S_h(0)$ in Fig. 1), the Hippocampus rapidly captures its memory trace [17]. The strength of this trace decays (is forgotten) over time at a rate D_h. Simultaneously, information is transferred to the Neocortex at a rate C (consolidation rate), converting it into a long-term memory with strength $S_c(0)$. The memory trace in the Neocortex also decays at a slower rate D_c. This "fast learning-slow consolidation" complementary mechanism [18,19] ensures that important information is retained long-term ($S_c(0)$ remains high), while non-critical information is filtered and forgotten ($S_h(0)$ decays rapidly), essentially solving the catastrophic forgetting problem in biological intelligence.

2.2. CLS-inspired mapping for machine learning models

Inspired by CLS theory, this study constructs a rigorous mapping framework to translate biological memory mechanisms into core components of machine learning models, aiming to endow models with anti-forgetting continual learning capabilities, as shown in Table 1.

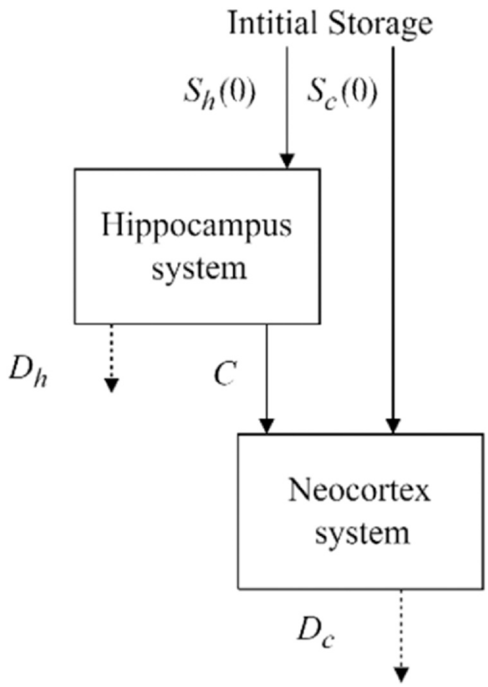

Fig. 1. Complementary learning system [11].

3. Online monitoring problem analysis and motivation

3.1. False alarms induced by distribution shift

A sample is identified as abnormal when either the T^2 or Q statistic exceeds its respective threshold. Traditional PCA condition monitoring models operate by converting real-time data into T^2 and Q statistics, which are then compared against corresponding thresholds. A sample is identified as abnormal when either the T^2 or Q statistic exceeds its respective threshold. Detailed descriptions of PCA and DPCA can be found in Refs. [20–22]. As shown in Fig. 2, normal data should fall within the critical region (yellow area), while abnormal data exceed the thresholds (red area).

The critical value for T^2 is determined by the significance level α. For example, when $\alpha = 0.01$, data points whose T^2 value exceeds this critical value are considered abnormal at the 1 % significance level. Traditional PCA models use a fixed training set, resulting in fixed thresholds that become increasingly mismatched as operating conditions evolve. Analysis of PCA models reveals that data identified as abnormal can be categorized into three types:

1) Genuinely abnormal data;
2) normal data affected by noise or disturbances;
3) normal data exhibiting distribution shifts during operation.

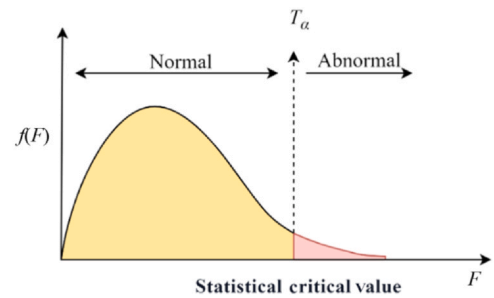

Fig. 2. F distribution for threshold calculation.

In long-term operation, the third type—normal data with distribution shifts—becomes increasingly significant, leading to excessive false alarms. This phenomenon is known as covariate shift, occurring when the operational data distribution gradually deviates from the training data distribution.

3.2. Limitations of dynamic PCA

To address this issue, the traditional approach involves establishing a DPCA model. DPCA performs well for short-term time-series monitoring data because it constructs an augmented matrix incorporating both training data and real-time monitoring data. The time-lag factor s within the augmented matrix can cover temporal variations to some extent. However, in long-term power plant operation monitoring, the deviation between real-time monitoring data and training set data grows increasingly large over time, exceeding the margin covered by the time-lag factor s. Consequently, DPCA also eventually generates a large number of false alarms, as illustrated in Fig. 3.

3.3. Trade-off between training set size and monitoring capability

Increasing the time span of the training set can enhance model robustness. As shown in Fig. 4, when the training set to test set ratio reaches 1:1 (DPCA) or 3:1 (PCA), false alarms are significantly suppressed. This is because the expanded training set has greater variance, enabling it to cover the distribution characteristics of subsequent monitoring data.

However, expanding the training set size also reduces the model's sensitivity to early-stage faults (Fig. 5). PCA and DPCA monitoring results statistics (sampling points) are shown in Table 2.

Short training set. Low fault detection delay (PCA/DPCA: 0 sampling points), but high false alarm rate.

Long training set. False alarm rate approaches zero (0 for both PCA and DPCA), but fault detection delay increases significantly.

This phenomenon aligns with the Receiver Operating Characteristic (ROC) trade-off for multivariate alarms: false positive rate and false negative rate (missed detections) exhibit a trade-off relationship.

Table 1
Concept mapping relationship between biological systems and machine learning.

Biological concept	Machine learning mapping	Operational definition and role
Hippocampus	Short-term memory buffer	The real-time input data stream x_t. Analogous to transient perception, only retains unprocessed new information.
Neocortex	Long-term memory bank (training set X_{train})	Stores refined core knowledge. The source of model skills (performance) determines the PCA loading matrix $\mathbf{P}$ and statistic thresholds.
Consolidation process (C)	Learning gate mechanism (L_t)	Controls the conditions for migrating valuable data from the short-term buffer to the training set X_{train}.
Neocortical forgetting (D_c)	Forgetting gate mechanism (F_t) Timeliness (w)	Identifies and removes obsolete or redundant samples from the training set X_{train}, preventing knowledge rigidity.
Memory trace strength ($S(t)$)	Data timeliness weight	Learning factor $S_l(t)$ and forgetting factor $S_f(t)$, characterizing the novelty/importance of real-time data, controlling the frequency of data migration and removal.

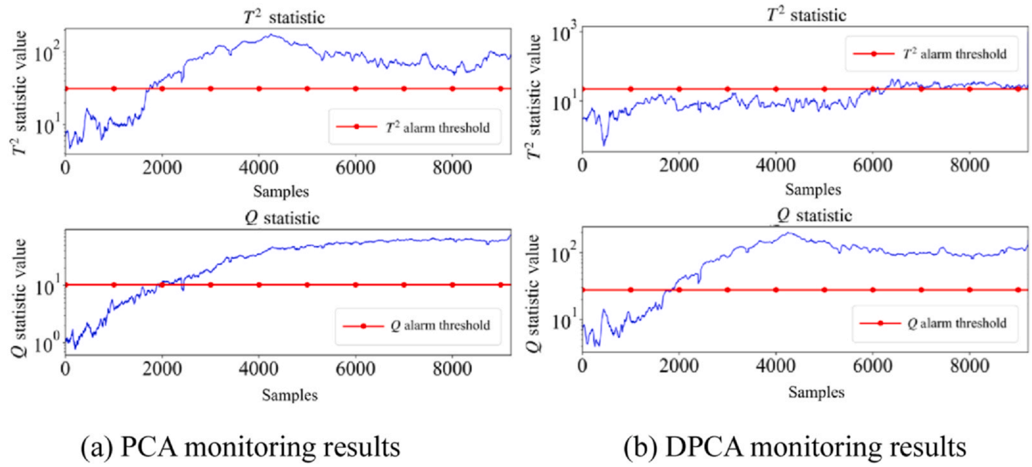

Fig. 3. PCA and DPCA monitoring results on long-period operating power plant data.

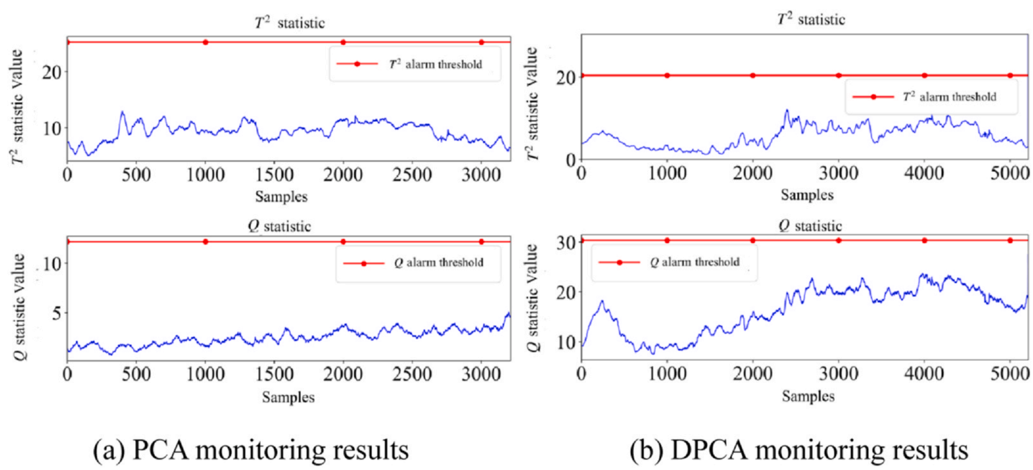

Fig. 4. PCA and DPCA monitoring results after adding training set samples.

Essentially, a long training set approximates the "global distribution" of monitoring data with high variance, reducing sensitivity to anomalies in specific periods. Conversely, a short training set maintains high consistency with the current data distribution but is susceptible to local fluctuations. DPCA, by introducing extra variables through the augmented matrix, further reduces false alarms but also prolongs fault response time (DPCA delay is higher than PCA in Table 2).

While offline scenarios allow performance optimization by adjusting training set size, maintaining consistency between the training set and real-time data is challenging in real-time monitoring due to the dynamic evolution of data distributions. Therefore, CLS-PCA introduces a learning/forgetting mechanism to dynamically update the training set to track data distribution drift, thereby simultaneously reducing both false alarm rates and missed detection rates.

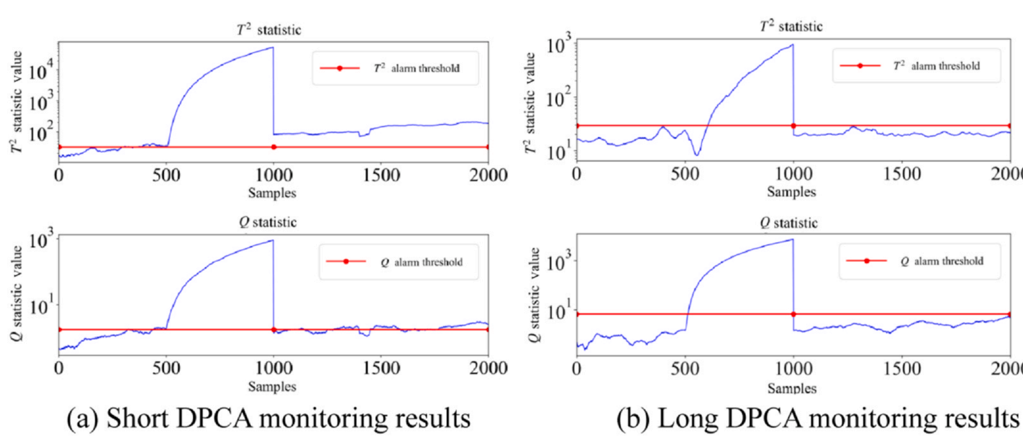

Fig. 5. Short and long DPCA monitoring results for a 5 % fault.

Table 2
PCA and DPCA monitoring results statistics (sampling points).

Fault type	Method	Training set	T^2 statistic	Q statistic
False alarms	PCA	Short	1524	5946
	PCA	Long	0	0
	DPCA	Short	5960	1784
	DPCA	Long	0	0
Fault detection delay	PCA	Short	0	0
	PCA	Long	105	14
	DPCA	Short	264	31
	DPCA	Long	284	39

4. CLS-PCA: methodology and algorithmic design

4.1. Problem background

Consider a condition monitoring system processing multivariate sensor data $X \in R^\wedge(n \times m)$ from a power plant, where n is the number of samples and m is the number of sensors. Traditional PCA models assume the statistical properties of X remain stationary over time. However, during long-term operation, data distribution gradually shifts due to component aging, operational changes, and environmental factors.

Let $P_t(X)$ denote the probability distribution of sensor data at time t. Under ideal conditions, $P_t(X) = P_0(X)$ for all t. In reality, as t increases, $P_t(X)$ deviates from $P_0(X)$, causing PCA models trained on initial data to generate increasingly more false alarms. Our objective is to develop a monitoring framework that maintains model relevance by adapting to the evolving data distribution $P_t(X)$, while preserving previously learned normal operating patterns.

4.2. CLS-inspired memory architecture design

The CLS-PCA model draws on the CLS theory, mapping the data flow in process monitoring to a memory process, as shown in Fig. 6: real-time monitoring data corresponds to instantaneous memory; the model's process of calculating statistics and making decisions based on current data corresponds to short-term memory; storing historical data used to build the PCA monitoring model corresponds to long-term memory.

Short-term memory vuffer (Hippocampal system). This component temporarily stores recent operational data (x_t, x_{t-1}, $...x_{t-w}$), where w is the window size. Unlike the biological hippocampus, our buffer has a fixed capacity and operates based on a sliding window.

Long-term knowledge base (Neocortical system). This corresponds to the PCA model's training dataset $D = \{x_1, x_2, ..., x_n\}$, which encodes stable patterns of normal operation.

The core innovation of CLS-PCA, distinguishing it from classical methods (like fixed window updates) and LSTM (which focuses on gating information flow), lies in introducing a controlled, active learning and forgetting mechanism to achieve dynamic updates of long-term memory (the training set). This mechanism relies on two key factors and corresponding gates.

1) Learning factor $S_l(t)$. Quantifies the value or confidence level of new knowledge (e.g., new operating conditions, new feature patterns) contained within the current real-time data.

2) Learning gate L_t. Activated ($L_t = 1$) when $S_l(t)$ exceeds a preset threshold, triggering the integration of the current data (or its key features) into the training set.

3) Forgetting factor $S_f(t)$. Quantifies the relative value or obsolescence degree of certain old knowledge (e.g., outdated operating conditions, redundant or no longer occurring data patterns) within the training set.

4) Forgetting gate F_t. Activated ($F_t = 1$) when $S_f(t)$ exceeds a preset threshold, triggering the removal of specific old data from the training set.

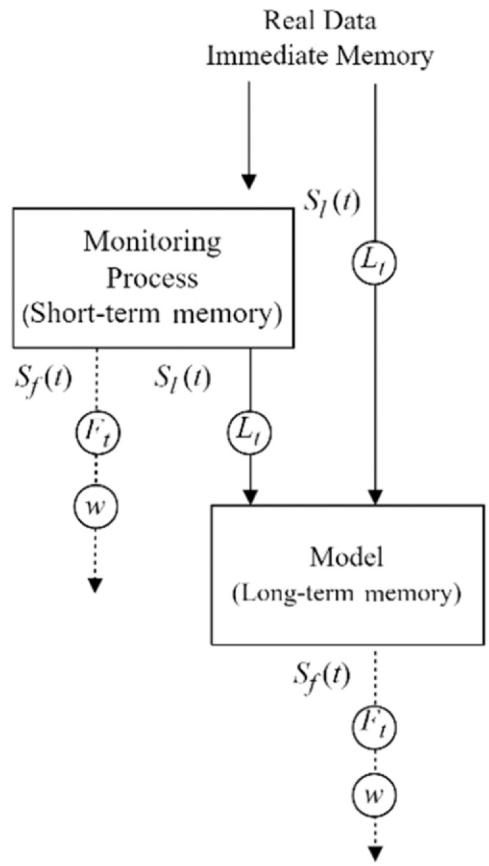

Fig. 6. CLS-inspired memory architecture.

$$S_l(t) = \begin{cases} 1, & T^2(Q) \text{ is always higher than } l_l^T(l_l^Q) \text{ in } t_1 \\ 0, & \text{other} \end{cases} \tag{1}$$

$$S_f(t) = \begin{cases} 1, & T^2(Q) \text{ is always lower than } l_l^T(l_l^Q) \text{ in } t_2 \\ 0, & \text{other} \end{cases} \tag{2}$$

The value of t_1,t_2 is usually greater than that of s, where s represents the time lag factor and reflects the overall volatility of the data. The specific value of t_1,t_2 must be determined based on the data conditions and the result of the s calculation.

$$l_{\text{new}}(s) = l(s) - \sum_i^{s-1} (s - i + 1) l_{\text{new}}(i) \tag{3}$$

In Eq. (3), when $s = 0$, $l_{\text{new}}(s) = l(s)$, where $l(s)$ is the number of variables minus the number of principal components. The formula is then recursively calculated until $l(s) < 0$.

The design of $S_l(t)$ and $S_f(t)$ is the cornerstone of model stability. They act as “intelligent filters” for update decisions, ensuring that learning operations occur only when significant and reliable new information is detected, and forgetting operations occur only when genuinely obsolete or redundant old information is identified. This effectively suppresses excessive updates caused by data noise or minor fluctuations, ensuring model robustness in dynamic environments. Crucially, the learning and forgetting processes for the T^2 and Q statistics are entirely independent, each possessing its own dedicated $S_l(t)$, $S_f(t)$, L_t, and F_t. This allows the model to adaptively adjust to changes in different types of anomalous features.

The long-term memory update process strictly follows cognitive logic: instantaneous memory (real-time data) cannot directly enter long-term memory (training set). It must first participate in forming short-term memory (internal state calculation), and whether the knowledge it contains can be consolidated depends on the evaluation

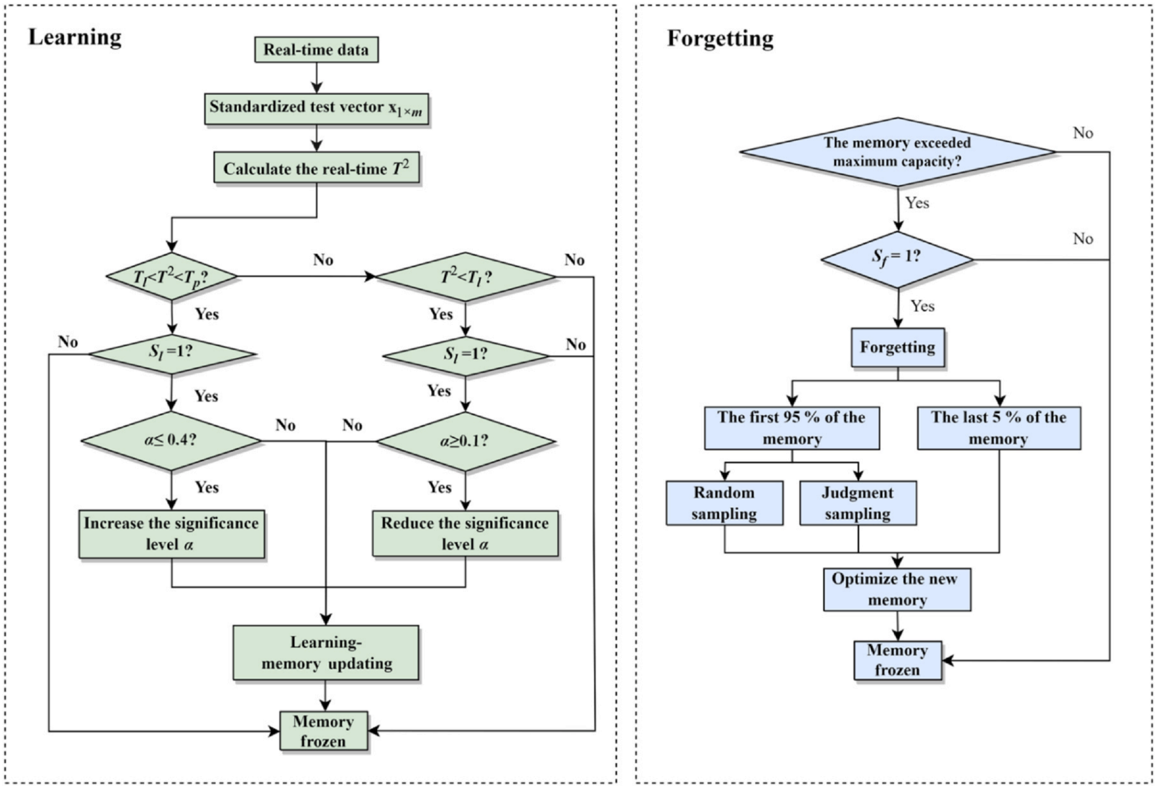

Fig. 7. Learning and forgetting flowcharts.

by the learning factor $S_l(t)$ and the triggering of the learning gate L_t. Similarly, knowledge in long-term memory does not automatically decay; its forgetting must be triggered by the evaluation of the forgetting factor $S_f(t)$ and the triggering of the forgetting gate F_t. This controlled mechanism simulates the selective process of human memory consolidation and decay. For specific model update and forgetting procedures, refer to Ref. [23].

4.3. Learning and forgetting mechanisms

The flowchart of learning and forgetting is shown in Fig. 7. Taking the T^2 statistic as an example (the same applies to the Q statistic), three T^2 thresholds are used to balance model stability and adaptability by dividing incoming data into four categories:

Fault data ($x > T^2_{f=0.01}$). Treated as abnormal and excluded from learning.

Risky data ($T^2_{p=0.05} < x < T^2_{f=0.01}$). Potentially unstable; not learned to preserve model integrity.

Learnable data ($T^2_{l=0.1} < x < T^2_{p=0.05}$). Reflects meaningful variation; eligible to be added to the training set.

Mastered data ($x \leq T^2_{l=0.1}$). Redundant; excluded to avoid overfitting.

Learning is further regulated by stability indicators derived from $S_l(t)$ or $S_f(t)$. If the T^2 statistic remains below the learnable threshold $T^2_{l=0.1}$ for a duration t_1, the model is considered stable; the learning threshold $T^2_{l=0.1}$ is relaxed (by increasing the significance level α associated with its calculation) to reduce unnecessary updates. Conversely, if T^2 exceeds $T^2_{l=0.1}$ for a duration t_2, instability is detected, and α is decreased to enhance learning (making the threshold stricter, allowing more data to be considered learnable. $\alpha \in (0.1, 0.4)$, the step size is set to 0.005).

The thresholds are determined based on the statistical confidence level (α) associated with the T^2 and Q statistics, and are dynamically adjusted according to the model stability metric (the duration of time the model remains above/below the learning threshold). The initial values of these parameters are empirically set based on stability requirements and will be adaptively adjusted based on the data. This adaptive mechanism ensures timely model updates with informative data while suppressing redundancy during stable operation, achieving effective learning and forgetting.

5. Case study

5.1. Experimental setup

5.1.1. Dataset

We evaluated CLS-PCA using 12 months of real operational data from a power plant. The dataset includes 10000 samples and 30 sensor measurements, capturing normal operation and various fault scenarios. Data from the first month (1000 samples) served as the initial training set, while the remaining 11 months (8914 samples) constituted the test set.

5.1.2. Baseline methods and evaluation metrics

We compared CLS-PCA with two baseline methods.

1) Standard PCA: Fixed training set with traditional threshold calculation.
2) DPCA: Utilized a time-lagged augmented matrix with $s = 5$.

All models were initialized with the same 1000-sample initial training set for fair comparison.

We employed two primary evaluation metrics.

1) Number of false alarms. Count of normal samples incorrectly classified as abnormal.
2) Number of missed detections (false negatives). Count of abnormal samples incorrectly classified as normal.

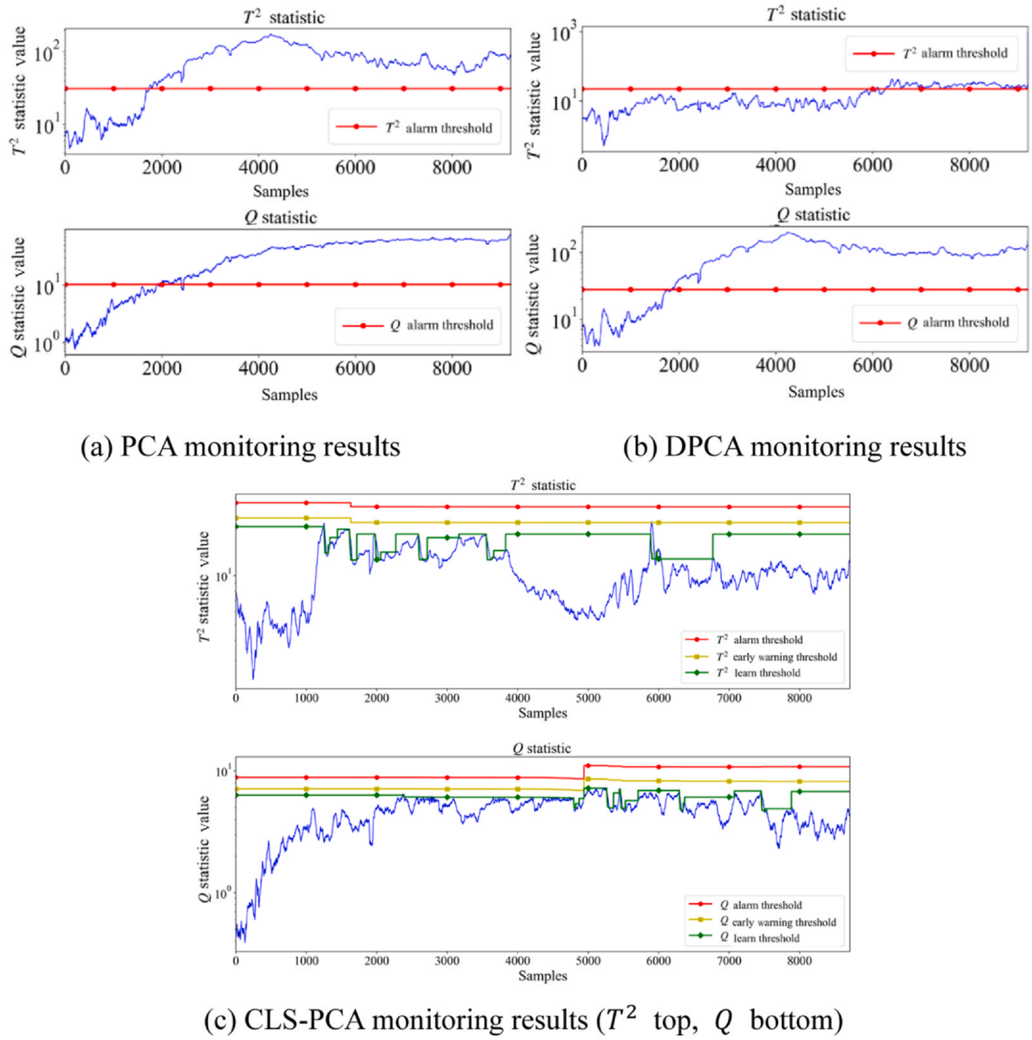

Fig. 8. Monitoring results of PCA, DPCA, and CLS-PCA on long-term operation data.

5.2. Long-term operational data analysis

The significance level α of the CLS-PCA model adaptively updates to adjust the model's learning rate. In Fig. 8(c), the red, yellow, and green curves represent three distinct thresholds. The blue curve represents the real-time monitoring data. As shown in Fig. 8(c), the thresholds in the CLS-PCA monitoring results adaptively change based on data variations. Data exceeding the red threshold is considered faulty, data between the yellow and red thresholds is considered risky (warning), data between the green and yellow thresholds is learnable normal data, and data below the green threshold is mastered (redundant) data. After model learning or forgetting, all thresholds (learn, warning, alarm) change. However, updates to α only affect the learn threshold (green curve). If only the learn threshold changes in Fig. 8(c), it indicates an α update without a full model update. If all three thresholds (learn, warning, alarm) change simultaneously, it signifies a full model update.

This section tests CLS-PCA on long-term operational data from a power plant to validate its adaptability to long-term data and the effectiveness of its adaptive thresholds. We compare PCA, DPCA, and CLS-PCA. The test results are shown in Fig. 8. As seen in Fig. 8(a) and (b), when the training dataset is small, PCA and DPCA monitoring performance is poor, generating numerous false alarms. In contrast, CLS-PCA effectively suppresses false alarms caused by long-term operation through continuous learning and forgetting.

Fig. 8(c) shows that the three thresholds for both the T^2 and Q statistics in CLS-PCA exhibit dynamic changes, generally trending downward. In the early monitoring phase (before sample point 2000), the real-time curves for T^2 and Q statistics show an upward trend, indicating that the monitoring data is gradually deviating from the training data. The three T^2 thresholds of CLS-PCA drop simultaneously around sample point 2000, indicating a full model update specific to the T^2 statistic. The three Q statistic thresholds drop simultaneously around sample point 1800, indicating a full model update specific to the Q statistic. Subsequently, the Q statistic remains relatively stable without significant changes. Minor threshold changes occur between sample points 5000 and 6000, suggesting that after the first update, the Q statistic has largely adapted to the current monitoring data distribution and no longer requires substantial learning or forgetting of the training set.

The updates for the T^2 and Q statistics are mutually independent. The T^2 statistic undergoes more frequent updates during the mid-monitoring phase (sample points 2000–4000). Frequent changes in the T^2 learn threshold signify frequent α updates, indicating the model is continuously adjusting its learning rate for T^2. Subsequently, all three T^2 thresholds decrease simultaneously around sample point 3850,

Table 3
False alarm counts for PCA, DPCA, and CLS-PCA monitoring.

Method	T^2 statistic	Q statistic
PCA	1524	5946
DPCA	5960	1784
CLS-PCA	1	0

representing another full model update. Starting from sample point 4000, through learning and forgetting, the T^2 statistic gradually adapts to the current operational data. The decreasing trend of the T^2 real-time curve indicates that the CLS-PCA model's match with the real-time data is continuously improving.

According to the statistical results in Table 3, the CLS-PCA model produced only 1 false alarm during monitoring. In contrast, PCA and DPCA models generated significantly more false alarms. This test demonstrates that CLS-PCA possesses superior capability in suppressing false alarms during long-term monitoring compared to traditional PCA and DPCA models, making it more suitable for industrial monitoring environments characterized by long operational cycles or strong slow time-varying data effects.

5.3. Fault detection performance

Tank leakage is one potential fault during power plant operation. Volume Control Tank (VCT) leakage has a relatively small impact scope and does not cause widespread parameter abnormalities like a severe accident. Furthermore, some parameters exhibit slow-changing trends after VCT leakage. This section uses a simulated VCT leakage fault to test whether the CLS-PCA model mistakenly incorporates slowly changing fault data into the training set, thereby rendering it unable to detect the VCT leakage fault. The fault was injected into the simulator at the 5th sample point to test the model's detection capability. Results are shown in Fig. 9. Due to the low fault magnitude and limited affected parameters, model detection exhibits a delay.

From Fig. 9, both T^2 and Q statistics show an upward trend. Crucially, the CLS-PCA model did not learn the fault data as new knowledge. This is because, according to the L_t classification, only data between the learn threshold T_l and the warning threshold T_p (green and yellow thresholds) is eligible to be saved as short-term memory. However, in Fig. 9, data within this learnable region persisted for only 1 s. Since the model's minimum learning time interval is 3 s, this portion of fault data was not transferred to long-term memory for storage. The reason this fault data was not mistakenly learned is that its duration within the learn threshold interval was insufficient. Consequently, the model did not incorporate this fault data, nor did it perform any model updates or α updates during this fault period. Fig. 10 shows a partial enlarged view of the CLS-PCA monitoring results (logarithmic axis) near the fault onset, confirming the fault data points fall below the learn threshold. The number of missed detections for each method is shown in Table 4.

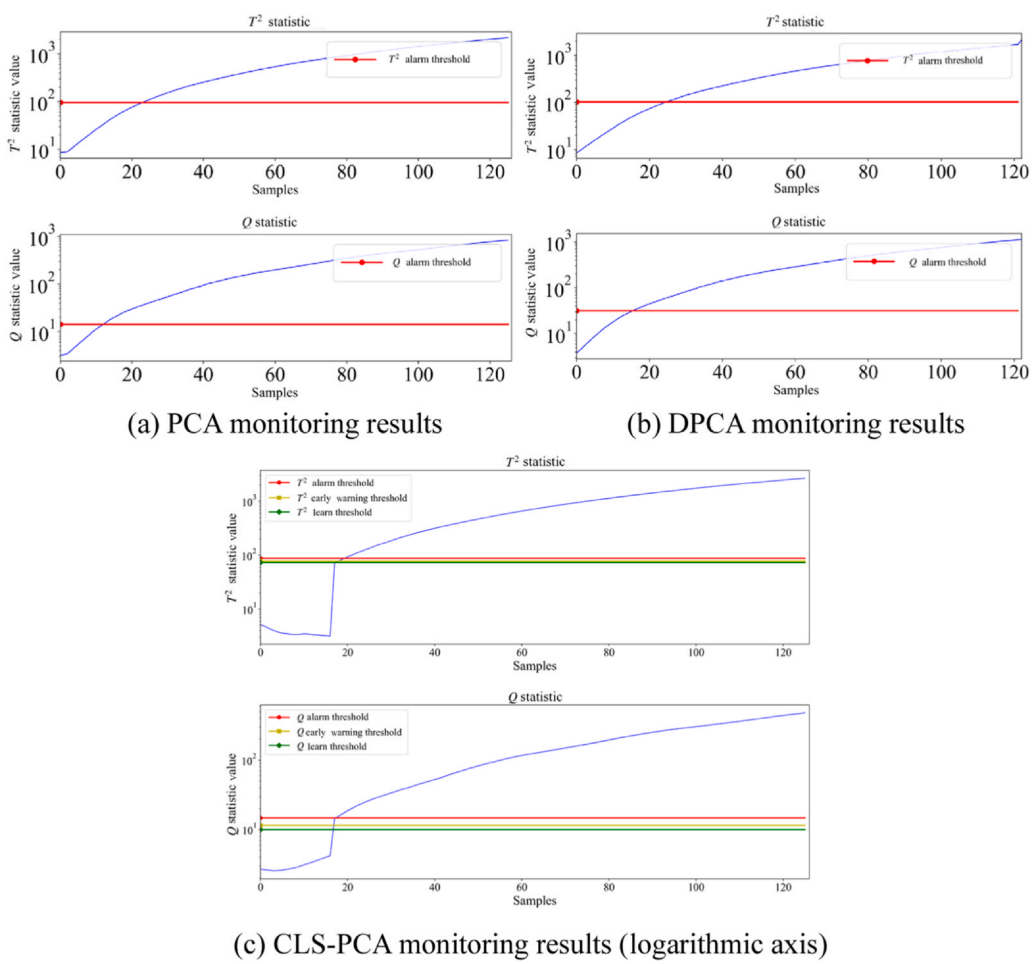

Fig. 9. Monitoring results of PCA, DPCA, and CLS-PCA for VCT leakage fault (logarithmic axis).

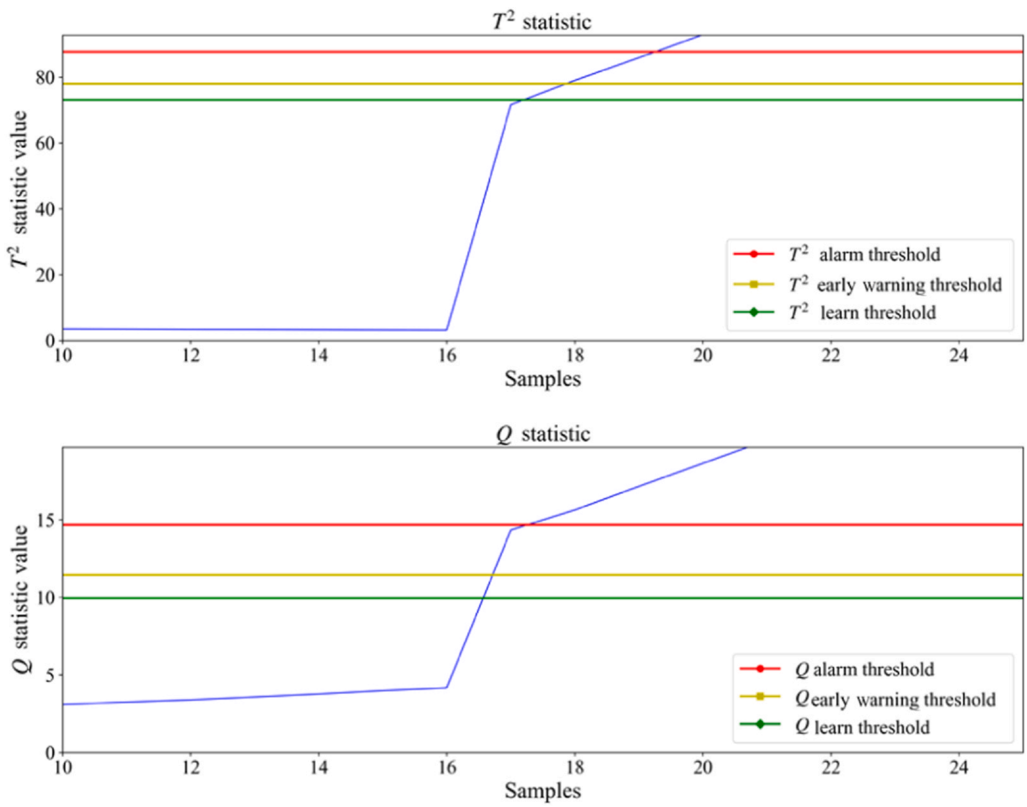

Fig. 10. A partial enlarged view of the monitoring results of the CLS-PCA model.

Table 4
Missed detection counts for PCA, DPCA, and CLS-PCA monitoring.

Method	T^2 statistic	Q satistic
PCA	21	9
DPCA	20	13
CLS-PCA	21	8

6. Conclusion

This paper proposed a novel condition monitoring method, CLS-PCA, based on the CLS theory, effectively addressing the challenges of catastrophic forgetting and rising false alarm rates faced by traditional PCA models in long-term operation systems like power plants. By introducing the hippocampal-neocortical complementary learning mechanism from neuroscience into condition monitoring, we designed a biologically plausible learning and forgetting mechanism enabling dynamic management of the PCA model's training set. The CLS-PCA model, through covariance decomposition and low-dimensional mapping techniques combined with adaptive threshold calculation, allows the model to continuously adapt to slowly evolving operating conditions while maintaining high fault detection capability. Experimental results demonstrate that compared to traditional PCA and DPCA methods, CLS-PCA significantly reduces the number of false alarms in long-term operational data monitoring and avoids mistakenly incorporating slowly changing fault data into the training set, thereby preserving detection capability for minor faults.

The primary innovation of this research lies in establishing a bridge between neuroscience theory and industrial condition monitoring, proposing a dynamic model update mechanism with clear biological inspiration. However, the method may face challenges in handling abrupt data changes, and optimizing model parameters (e.g., learning factor, forgetting factor) still relies on domain knowledge. Future work will focus on three directions: 1) developing parameter self-adaptation strategies to reduce dependence on manual experience; 2) extending the CLS theory to deep learning frameworks to enhance recognition capability for complex fault patterns; 3) exploring the applicability of this method in more industrial scenarios (e.g., wind power, chemical processes) to validate its universality. Moreover, within the domain of rail transit and associated heavy-duty applications, such leakage-type faults are of particular pertinence. For instance, fuel tank leakage in mining dump trucks or coolant leakage in traction systems exhibit similar slow and progressive characteristics. Consequently, the proposed CLS-PCA mechanism can be directly extended to these domains without modification, providing valuable insights for condition monitoring in broader transportation and industrial systems.

CRediT authorship contribution statement

Yue Yu: Writing – review & editing, Writing – original draft, Methodology, Data curation. **Jianhui Yi:** Conceptualization. **Deshui Han:** Investigation, Funding acquisition, Formal analysis. **Xinkang Li:** Visualization, Validation, Software. **Xiaohui Zhang:** Supervision, Resources, Project administration. **Zeyun Yang:** Methodology, Funding acquisition, Conceptualization.

Data availability

The authors do not have permission to share data.

Declaration of Competing Interest

The authors have declared that they have no conflicts of interest relating to this work. We also declare that we have no commercial or associative interests that could potentially lead to conflicts of interest with the work submitted. Furthermore, we assert that there are no conflicts of interest regarding the publication of this paper.

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